

Using the whole SD card

This may sound a bit weird, but by default, the Raspberry Pi only uses as much of the SD card as the operating system requires. This means that even though you might have used a larger SD card, all its resources won't be used. To fix this, so that all the space on the SD card can be used, use the up/down cursor keys to select the '*expand_rootfs*' menu option and hit *Enter*. Once you do that, a script will run and you will get the following confirmation:

Using Full Screen of the monitor:

Raspberry Pi, by default, doesn't use the full screen of your monitor; it just appears in the middle portion of the screen. This is not true of all monitors, but if it happens, then selecting the option to *Disable Overscan* may fix this for you. Use the left and right cursor keys to make your selection and then hit *Return*. If after disabling Overscan, you cannot see the left edge of the screen, then check the section '*Running Raspi_Config After Booting*'.

Booting into the desktop

Raspberry Pi, by default, boots you into the command line interface. So, to boot into the desktop, select the *Boot_behaviour* option; then, select '*Yes*' and press *Enter*.

Testing the Internet connection

Using the Ethernet cable: The fastest way to get your Raspberry Pi connected is to use an Ethernet patch cable and just plug it into the back of your router. As soon as you plug your Pi in, you should see the network LEDs start to glow. If that doesn't happen, it means your router is not configured to DHCP (Dynamic Host Configuration Protocol). You can fix that by going to your router's settings in the computer to which it is already connected, and enabling the option that configures DHCP for your router.

Setting up Wi-Fi using the command line: Open the terminal, and use the following commands:

```
sudo nano /etc/network/interfaces

auto lo
iface lo inet loopback
iface eth0 inet dhcp
allow-hotplug wlan0
auto wlan0
iface wlan0 inet dhcp
    wpa-ssid "your_ssid"
    wpa-psk "your_password"
```

You have to make changes in the last two lines, changing *your_ssid* and *your_password* to the passwords of your Wi-Fi connection and SSID, respectively. After changing that, press *Ctrl + x* to save the file. Restart your Raspberry Pi and

```
pi@raspberrypi:~$ sudo apt-get install tor
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following extra packages will be installed:
  tor-geoipdb torsocks
Suggested packages:
  miniserver aut-ext-torbutton socat tor-arm polipo privacy apparmor-utils
The following NEW packages will be installed:
  tor tor-geoipdb torsocks
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 2,494 kB of archives.
After this operation, 7,386 kB of additional disk space will be used.
Do you want to continue [Y/n]? Y
Get:1 http://mirrorsirector.raspbian.org/raspbian/ wheezy/main tor armhf 0.2.3.1-1 [1,168 kB]
Get:2 http://mirrorsdirector.raspbian.org/raspbian/ wheezy/main torsocks armhf 1:1-3 [716 B]
Get:3 http://mirrorsdirector.raspbian.org/raspbian/ wheezy/main tor-geoipdb all 2.2-2-1 [1,412 kB]
Fetched 2,494 kB in 17s (354 kB/s)
Selecting previously unselected package tor.
(Reading database ... 62860 files and directories currently installed.)
Preparing tor (from .../tor_0.2.3-25-1_armhf.deb) ...
```

Figure 1: Installing TOR

you will see it connect to your Wi-Fi as it boots up.

To find the IP address of your Raspberry Pi, open the terminal and type the following command:

```
sudo ifconfig
```

Next, at the *wlan0*, you will see *inet address*, which is your Pi's IP address.

Installing Tor

Now, let's install TOR—the onion routing software—step by step. Log in to your Raspberry Pi by Ethernet or Wi-Fi, and open the terminal:

```
sudo apt-get update
sudo apt-get install tor
```

Now edit the TOR config file by running the following command:

```
sudo nano /etc/tor/torrc
```

Next, write the following code to it, right below the FAQ option:

```
Log notice file /var/log/tor/notices.log
VirtualAddrNetwork 10.192.0.0/10
AutomapHostsSuffixes .onion,.exit
AutomapHostsOnResolve 1
TransPort 9040
TransListenAddress 192.168.42.1
DNSPort 53
DNSListenAddress 192.168.42.1

TORConfig.jpg
```

Now, change the host access point to something else, which we'll call Onion Pi for now. Don't forget to set a good password—you should not use the default option here!

```
sudo nano /etc/hostapd/hostapd.conf
```